

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3
01.2	<i>1 mark for each point.</i> • Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
01.3	<i>1 mark for each point.</i> • Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) • Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels (1) - Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures (1) - Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x (1) - So natural fixation is sustainable indefinitely with a low carbon footprint (1)	4
02.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Hydrological cycle (evaporation, precipitation) (1) - Carbon cycle / photosynthesis (1) - Atmospheric circulation / winds (1) - Thermohaline circulation (density differences from heating / evaporation) (1) - Decomposition (1)	3
02.3	<i>1 mark for each point.</i> - Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting (1) - Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) (1) - Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
03.1	<i>1 mark for each point.</i> - Coordinating global data on biodiversity conservation (Red List) (1) - Increasing understanding of the importance of biodiversity (1) - Deploying nature-based solutions to global challenges in climate, food and sustainable development (1)	3
03.2	<i>1 mark for each point.</i> - Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ (1) - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km²) and adults (2100) only meet Endangered (1) - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered (1)	3
03.3	<i>1 mark for each point.</i> - Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed (1) - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) (1) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations (1)	3

Question	Marking guidance	Marks
03.4	1 mark. Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
04.1	1 mark for the correct answer. $1000 / 100\,000 \times 100 = 1\%$ (1)	1
04.2	1 mark for each point. - Some light is reflected by the leaves (1) - Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed (1) - Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) (1)	3
04.3	1 mark for each point. - Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) (1) - After four levels too little energy remains to support a further population (1)	2
04.4	1 mark for each point. - Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) (1) - Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat (1)	2
04.5	1 mark for each point. - Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) (1) - So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	1 mark for the correct answer. $2.7 - 2.45 = 0.25$ billion = 250 million years (1)	1
05.2	1 mark for each point. - For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans (1) - Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere (1)	2
05.3	Any 3 from the points below - 1 mark each, maximum 3. - Oxygen in the atmosphere absorbed ultraviolet light (1) - A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed (1) - Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) (1) - Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out (1)	3

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> • Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) (1) • The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) (1)	2
05.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion (1) • Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas (1) • Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
06.1	<i>1 mark for each point.</i> • About 40-50 traps (1) • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable (1)	2
06.2	<i>1 mark for each point.</i> • Widely scattered results (large standard deviation) mean a single sample may not be representative (1) • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed (1)	2
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Same type and size of pitfall trap, set for the same length of time (1) • Traps emptied at the same time of day / same weather conditions in both habitats (1) • Same number of traps and same method of positioning (random or systematic) in each habitat (1) • Other variables that could affect results (eg rainfall, temperature) monitored or controlled (1)	3
06.4	<i>1 mark for each point.</i> • Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally (1) • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends (1) • The interval between samples should match the rate at which the factor changes, found from a preliminary study (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
07.1	<i>1 mark for each point.</i> • Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable (1) • If it cannot be adjusted the percentage error should be recorded so the results can be corrected (1)	2

Question	Marking guidance	Marks
07.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (eg chewed hazel nuts): presence of species such as dormice (1) • Nests / burrows: presence and breeding (1) • Tracks / footprints: presence, movements (1) • Territorial marks (scratching posts): presence and territory (1)	3
07.3	<i>1 mark for each point.</i> • Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species (1) • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen (1)	2
07.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Marked individuals mix freely with the rest of the population (1) • Marks do not fall off or wear away, and marking does not affect survival (1) • The population does not change during the study (no births, deaths or migration) (1) • All individuals have an equal chance of being caught / animals are mobile and can be caught (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	<i>1 mark for each point.</i> • Very small gene pool so individuals are closely related and inbreeding is likely (1) • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low (1) • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) (1) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies (1)	4
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided (1) • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life (1) • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) (1) • Stress / disturbance (1)	3
08.3	<i>1 mark for each point.</i> • AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection (1) • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) (1) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
09.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hedgerows (1) • Hay meadows (1) • Grazed moorland / heathland (1) • Chalk grassland (1)	2
09.2	<i>1 mark for each point.</i> • Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive (1) • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers (1)	2
09.3	<i>1 mark.</i> • The introduced (non-indigenous) predator may eat other species as well as the pest (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9

Question	Marking guidance	Marks
10.2	1 mark. A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes - Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels - Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer - GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) - Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people - Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25